

❑ 14 Descriptive Statistics - Two variables

Consider the following questions.

1. Do diesel engines in cars have greater longevity than petrol engines?
2. Do dropout rates for first year student vary across the four faculties in a university?
3. Does educational attainment by women correlate with the number of children they bear?
4. Do average residential rents tend to be higher in more populous towns?
5. Are there differences between Waterford, Galway, and Limerick in the price of houses?
6. Does a novel weight-loss programme effective in reducing participant's weights?

To explore such questions using statistical methods we will need to identify the two variables and their type. The variables for the examples above are identified below in *italic* and their type is given in **bold**:

1. *longevity* (i.e., life span of the engine in years) - **continuous** and *fuel* (petrol or diesel) - **binary**
2. *faculty* (i.e., science, business, engineering, law) – **nominal** and *dropout* (yes or no) - **binary**
3. *education* (i.e., leaving cert, diploma, degree, masters) – **ordinal** and *children* – **discrete**.
4. *rent* (i.e., monthly average) – **continuous** and *population* (town population) – **continuous**.
5. *city* – nominal and *price* – **continuous**.
6. *before* weight (**continuous**) and *after* weight (**continuous**)

The first five of these questions (and a great many others than might arise in research) often amount to asking if there is a statistical association or relationship between the two variables. If there is an association, then knowing the value taken by one variable gives us *some* useful information as to what the value for the second variable might be.

In example 1, if an analysis suggests that diesel cars have a greater longevity then once we learn that the engine in a car is petrol we might expect a lower longevity, and conversely with a diesel car. I.e., if the datum taken by the binary variable *fuel* is “petrol” then our knowledge about what value the scale variable *longevity* might take is increased.

In example 2, if the drop-out rate is 17% overall but 9% for law student then we might adjust downwards our expectation that a random student will dropout if we know that they study law. Knowledge of the value taken by *faculty* gives us an idea as to the likelihood that the value taken by *dropout* might be “yes”.

In example 4, if we learn that residential rents tend to be higher in larger towns, then if we establish that a particular resident lives in a small village, we might expect the rent they pay to be lower. Again, knowledge of the value taken by *population* gives us some hint as to what the value taken by *rent* might be.

Example 6 differs from the others in that the question has a paired, or repeated measures nature to it. We have a pair of measures for each subject as we repeat the measure on each at two points in time.

As such, we could consider that the analysis involves a single variable obtained by subtracting the *after* weight from the *before*. If the mean of this new, derived variable is positive it suggests an average subject's weight loss, if it is negative, a weight gain and if it is zero, the weight loss programme has no impact.

The original variables (before and after weights) are plainly not independent – an individual of low/high *before* weight will tend to have a low/high *after* weight – we learn nothing useful from this. However, if the weight loss program is effective, and we know the *before* weight, we might anticipate that the *after* weight will not only be similar, but lower. This is the nature of “useful information” obtained due to association between paired variables.

N.B. In all problems where we explore statistical associations, we will be looking for tendencies rather than absolutes. For example, some residents in large towns may pay low rents level but that there might be one (or several!) cases in the data where town population is high but rent is low does not show that there is no association.

It is worth noting here the distinction between a statistical relationship and a mathematical one. The relationship between say petrol purchased at a service station and price paid is a mathematical one. Once we know how much petrol we buy we will have complete knowledge as to how much we must pay. This is not the case for the problems considered here, which involve statistical relationships, where we only have some, incomplete, knowledge of what our second variable will be once we learn the value of the first.

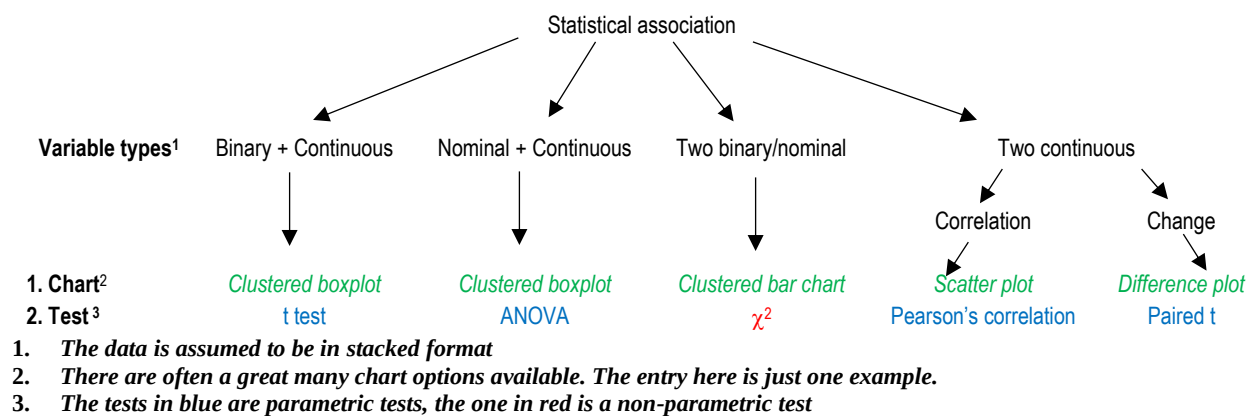
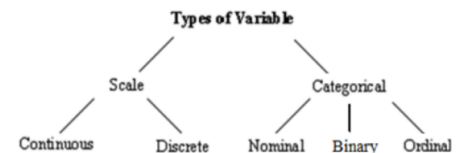
When exploring statistical relationships, using various methods, there are two questions we might pose.

1. What chart(s) might best be employed to visualize any possible (statistical) association?
2. What test might we use to decide if there is in fact an association?

Additionally, in some cases we might want to exploit the association between the variables to make predictions and quantify in some way our confidence in such predictions.

In this chapter we look at a number of problems involving possible associations between pairs of variables. Our focus here will be on selecting a suitable chart and identifying an appropriate statistical test. We will also consider elementary ways to make predictions. We will defer the details of how such statistical tests are applied and also, how we measure confidence in predictions until a later chapter.

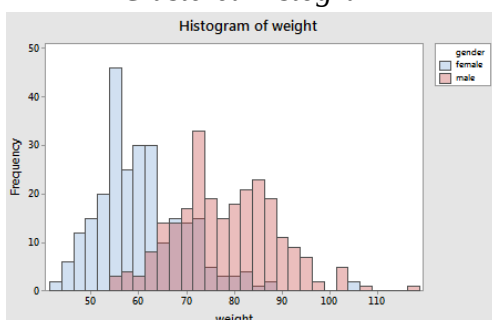
An overview of some of the basic methods and constructs in statistics that might be used to address these questions, given the types of the variables (for convenience, the categorisation we use is reproduced opposite) involved is outlined in the tree diagram below. Thus, being able to identify the variables involved, and their types is critical to deciding the analyses.



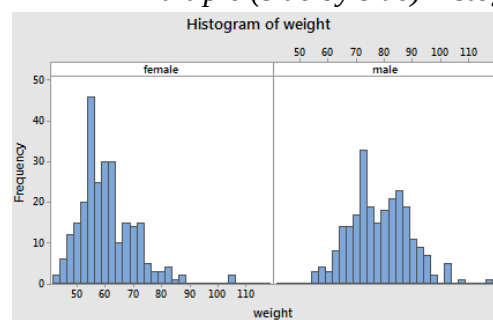
Clustered v multiple plots

It is important to note that different statistical software and textbooks may refer to the same chart type using different names. For the purpose of this module, and for charts of two or more variables, we make the following distinction between clustered and multiple charts. A multiple plot requires that the plotting region be duplicated, a clustered plot does not. Examples involving histograms are shown below, but the definition can also be applied to other chart types.

Clustered histogram



Multiple (side by side) histogram



Discrete and ordinal variables

Note that the diagram does not consider discrete (scale) or ordinal (categorical variables). We noted earlier than if variables of these type have a high granularity, then we can treat them as continuous and then use the tree diagram above.

If the granularity is low, then you might opt to ignore the variable's ordering and treat it as nominal. For example, smoking status (never < past < current) is an ordinal variable. If we were interested in exploring if there is a difference in lung function between the three groups then we could ignore this ordering and consider smoking status as consisting of three different, unrelated categories. Ignoring the ordering amounts to discarding information – so our analytical methods might not be optimal.

Independent and paired association

Students often have difficulty distinguishing between two independent samples and two paired samples investigations (e.g., t test v paired t test) as both are concerned with comparing the mean value between two groups.

This is less of a problem when the data is stacked as the types of the variables will be different (binary + continuous or continuous + continuous). However, if the data is unstacked then you will need to be able to distinguish between the two.

It should be noted that it would be sub-optimal rather than “wrong” to analyse data as independent samples when a paired samples analyses could be applied. But paired tests usually work better than independent samples tests and indeed, investigators will sometimes design their study in a particular way to induce a pairing in the data, as should be evident in some of the examples below.

For a paired analysis there is a link between the groups by virtue of the measure being taken on the same or similar subject or experimental unit.

Examples

1. Two electric drills might be compared by recording the time it takes to drill similar holes in masonry bricks with each brick drilled once with the first drill and once with the second.
2. The efficacy of two drugs might be compared by procuring a sample of patients and matching them in pairs on relevant criteria such as age, sex, smoking status and general health/fitness levels etc. and then proscribing one drug to one of the first of the pair and the other drug to the second.
3. The effectiveness of a weight loss programme might be assessed by weighing participants before the programme and then weighing the same participants after.
4. The difference between two time + temperature settings for baking scores on the calories / 100g might be determined by preparing batter mixes, making two scones from the mix and baking one each at the two separate settings.

For an independent samples analysis there is no such link between the groups.

Examples

1. Comparing drills where one set of bricks is used to test one drill and a different set to test the other.
2. The efficacy of two drugs where there is no matching of subjects between groups.
3. The longevity of cars depending on whether they have diesel or petrol engines.
4. The average earnings of male and female solicitors in a law firm.

• EDA

A formal approach to address the questions posed at the start of the chapter requires the constructs and models of inferential statistics (i.e., the tests stated in the tree diagram above). A less formal approach using exploratory data analysis (EDA) might be used to identify patterns in the sample data. And charts can be a very effective tool for this end.

We will return to the other aspects of investigating associations later – for the rest of this chapter we consider the business of selecting charts and use a number of examples to illustrate.

Scatter plot and correlation

Example

A student wishes to estimate their effective broadband speed. She obtains data on files she has downloaded – observing both the size of the file (Mb) and the download time (seconds). The data is tabulated below.

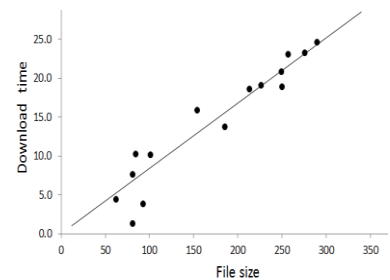
<i>File size</i>	93	101	62	249	250	154	290	185	257	276	226	84	81	81	213
<i>Download time</i>	3.9	10.2	4.4	20.9	18.9	15.9	24.7	13.8	23.1	23.3	19.1	10.3	1.3	7.7	18.6

We can readily see that the two variables here are both scale and that the question of interest is concerned with correlation between them. I.e., does one quantity (time) tend to increase or decrease in response to a change in a different quantity (size). A useful chart here is the *scatter plot*.

The scatter plot (Excel) is shown opposite.

As download time depends on file size (rather than the other way around) we would probably elect to plot size on the horizontal axis and download time on the vertical axis.

We can make predictions using the straight line shown (the regression line). For example, we might predict that a file size of 200 Mb will take 15 seconds to download.



A variation on the scatter plot is the marginal plot – an example is given in the [SETU retention](#) scenario.

The test we might apply to assess if the two variables are independent is a test that a quantity r called the correlation (Pearson's) differs from zero.

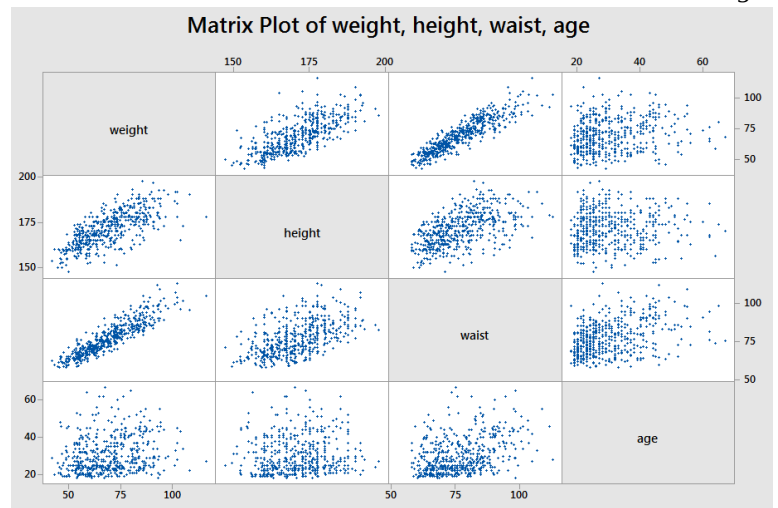
In many cases, including this one, it is abundantly clear without any tests, or even collected data, that the two variables would not be independent. It is plainly true that the greater a file size, the longer it would take to download.

Matrix plot

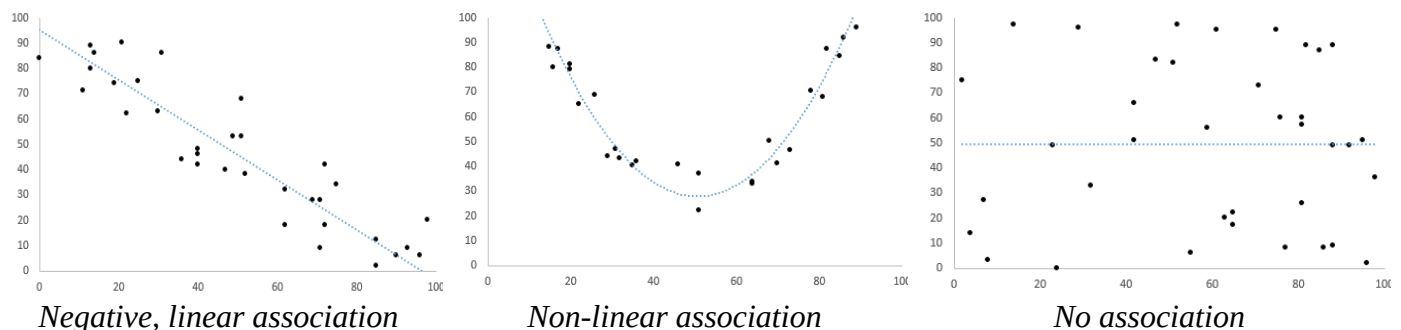
Sometimes, especially in studies in natural science or engineering, we might have quite a few scale variables. For example, in a sample of men we might record their weights, heights, ages and waist measurements. It might be of interest to see visually observe the relationship between these variables.

Rather than plot the variables in pairs we might use a statistical package such as Minitab to obtain a matrix plot – as shown opposite.

Can you see why waist measurement is the best indicator (or predictor) of weight and age is the worst?



Note that all example so far that showed any relationship showed a positive, linear relationship (e.g., as the X variables increases, so too did the Y variable). The association need not be linear, need not be positive or of course, need not exist at all - as illustrated in the examples below (all in Excel)



Difference plot and paired t test

Example

A group of 12 people undertake a novel, 6-week weight loss programme and it is required to produce a chart that shows the effectiveness of the programme. For each subject, their before and after weights are measured and from this, the difference (i.e., the weight loss) is calculated by subtraction.

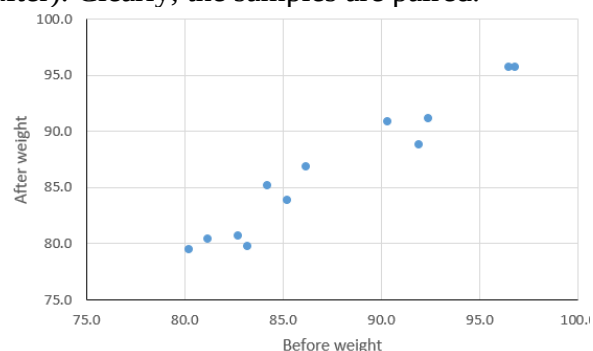
before	90.3	85.2	96.8	92.4	84.2	96.5	91.9	82.7	80.2	86.2	81.2	83.2
after	90.9	83.9	95.7	91.2	85.2	95.7	88.8	80.7	79.5	86.9	80.4	79.8
difference	-0.6	1.3	1.1	1.2	-1.0	0.8	3.1	2.0	0.7	-0.7	0.8	3.4

Again, the two variables here are both continuous but the question of interest now is concerned with whether there is a difference between two samples (before and after). Clearly, the samples are paired.

As we again have two scale variables, we could construct a scatter plot. But would this convey any useful information?

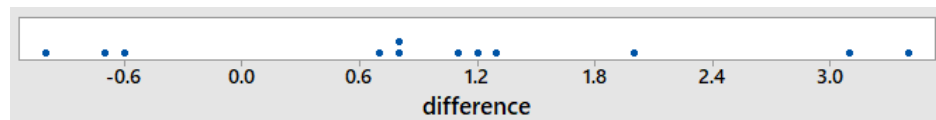
Such a plot (Excel) is produced opposite and readily shows that before and after weights are correlated.

But this is not very informative as it is informing us of something we already know – the heavier (lighter) subjects in the group before will tend to be the heavier (lighter) subjects in the group after.



The chart doesn't readily reveal whether *after* weights are lower than *before* weights.

A more useful plot might focus on the weight change, or difference from before to after – taking a dot plot, or boxplot or histogram of these values (we will call these difference plots). A dot plot (Minitab) is shown below.



Observe the distribution of the data relative to the value zero. Most of the values are positive indicating that most subjects lost weight. So, the plot suggests that the weight loss program is effective. If we calculate the average difference, we obtain 1.008 Kg. Thus, we might predict the weight loss attained by a new subject undertaking this weight loss programme to be about 1 Kg.

A suitable test here is the paired t test, which tests if the sample difference of 1.008 is different from zero.

The examples above should make clear that a correlation analysis might be applied when a paired test would make for a more meaningful analysis. What about the converse? Might a paired analysis be considered rather than a more appropriate correlation analysis?

Much of the time correlation is considered to explore association between two different quantities. E.g., do rent rates increase with town population, does alcohol consumption increase with age, does PC boot time decrease with amount of RAM. A paired test involves repeated measures on a subject, which implies that the same quantity (e.g., weight in the example above) is measured each time. If two different quantities are being measured, then it makes no sense to consider a paired samples analysis. It would be akin to asking – “Has my weight of 78 Kg this year changed from my height of 1.82 m last year?”

Clustered plot and t test

Example

A food processing firm packages dry pasta in packages of nominally, 500 g but the actual mass does vary. It is of interest to know if mass varies by production shift (morning or evening). Sample of products are taken from each shift and the mean calculated – the data is shown below.

Morning	494.2	497.0	496.1	499.8	496.5	500.3	498.6	496.4	497.5	499.8	501.1
Evening	503.6	497.4	502.3	505.5	500.8	499.6	503.1	500.2			

Note that in this problem we appear again to have two scale variables. But this is because our data is unstacked – i.e., samples are in different columns, or in this case rows

If we restructure our data in to stacked format as shown opposite, we have one scale and one binary variable.

Shift	Mass
Morning	494.2
Morning	497.0
Evening	503.6

We could calculate the mean mass for each shift and use this to make a prediction. For example, if the mean mass in the sample of products made in the morning is 499.3 then this would be our estimate for the mass of a product that is to be made in the morning shift.

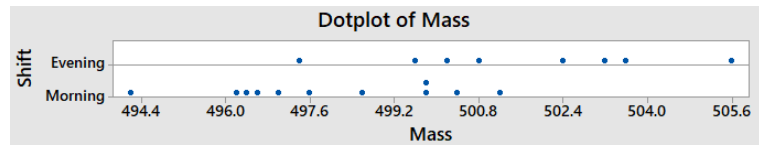
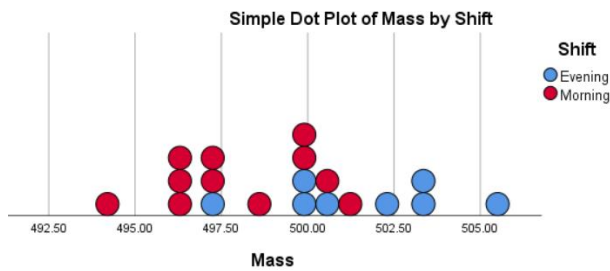
Thus, we have independent samples and a suitable test here is the t test, sometimes called the independent samples t test.

Note that we are the notion of independence in two different ways in this problem. We say the samples are independent samples because there is no pairing between them. We will also consider if the variables shift and mass are independent – if they are, then mass does not vary by line.

Our earlier tree diagram advises using clustered boxplots or clustered dot plots. But there are a good number of charts that are useful for visualising the data.

Clustered dot plot

As the samples are relatively small an effective chart to help assess any differences is the clustered dot plot. Such plots from both SPSS and Minitab are produced below.



Clustered dotplot (SPSS)

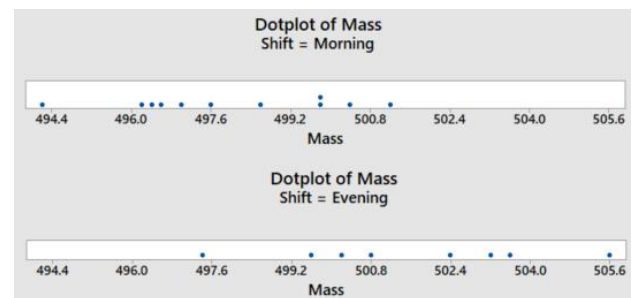
We describe the plot as clustered as we in chart data from both the morning and evening shift on the same plot. There is some evidence that the “evening” data is right shifted relative to the “morning” data, suggesting that the mass of packages produced on the evening shift may tend to be heavier.

Clustered dotplot (Minitab)

An alternative to the clustered-type plot is a what we will describe as a multiple plot – an example of which is shown opposite.

Multiple plots involve distinct charts and are aligned here to allow us to see, as with the clustered plot, that mass in the evening shift tends to be higher.

We will look at a number of variations on multiple and clustered plots below.

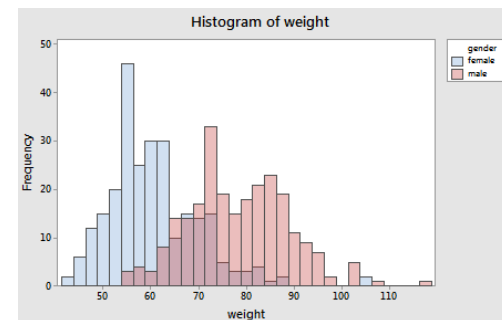


Clustered histogram

A data set is collected and records for a number of subjects, their weight and gender.

Opposite is constructed (Minitab) a single, clustered plot with separate histograms for males and females.

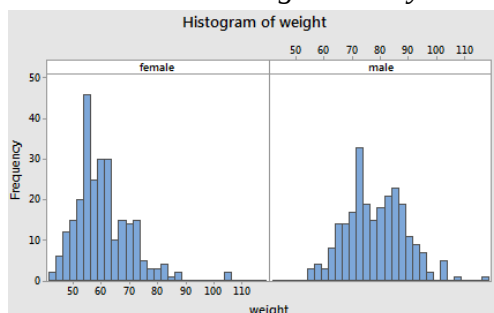
The chart for males is clearly right-shifted relative to the one for females – indicating (we would be expected) that males tend to be heavier.



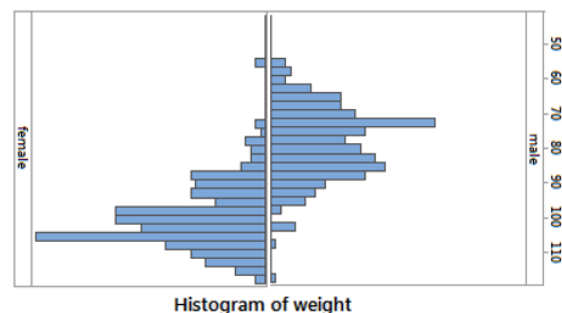
Multiple histograms (Side by side and back-to-back)

Below are multiple histogram plots for subjects' weight, by gender – both generated using Minitab.

Arranged side by side



Arranged “back-to-back”



Both the clustered histogram and the back-to-back histogram allow differences between the two to be more easily compared. The “male” plot is right shifted compared to the “female” plot, indicating that men weigh more. It is less easy to see this in a side-by-side histogram.

A variation of the “arranged in a column” histograms is available in SPSS, where rather than the one up, one down arrangement, both are oriented the same way.

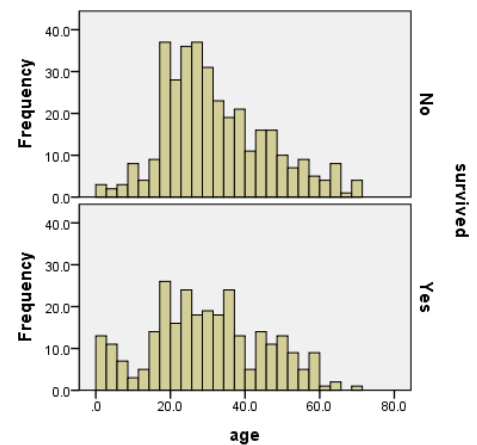
Example

One of the findings in relation to the titanic data is that the average (mean) age of the survivors was lower than for the non-survivors.

The multiple histogram (arranged in a column) opposite is generated in SPSS

However, there is no real indication that the “no” group is right shifted compared to the “yes” group.

Was the age profile of survivors really lower?



N.B. As you can see here, there are often a number of options available to plot your data. It is often a good idea to experiment with a number of different chart options - a pattern in a data set may not be especially evident in one chart but may be quite obvious in another.

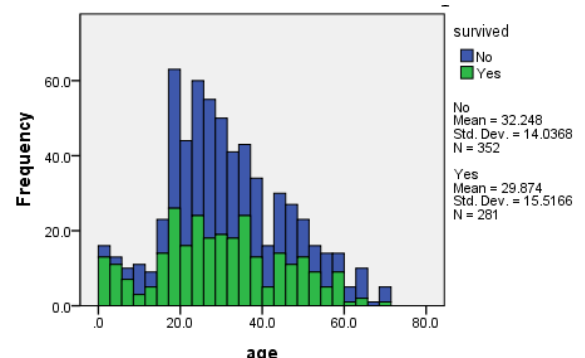
Stacked histogram

A stacked histogram is a type of clustered histogram with all the groups (two in the example here) represented by different colours / patterns but on one bar. This allows the breakdown of the groups within each bin on the histogram to be examined.

The survival by age data from the Titanic data set is shown opposite (SPSS) with some additional summary statistics.

Note the high concentration of green in the left most bars. These bars correspond to children and green represents the survivors. So, children had a higher propensity to survive.

And of course, children have lower ages bringing the average age of survivors down.



Had you only constructed one of the other variations of the multiple/clustered histogram you might have very well arrived at a spurious explanation as to why the age of survivors tended to be lower. E.g., younger passengers were more agile than older passengers and better able to secure a place on a lifeboat.

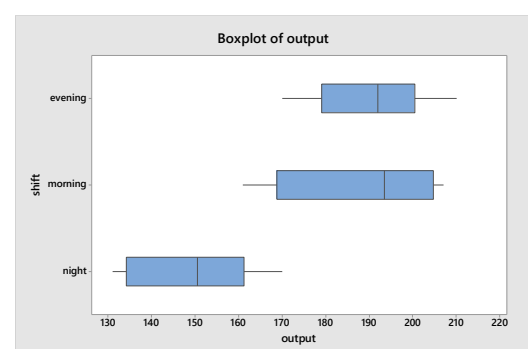
Side by side boxplot

Example

Output (total units manufactured) from a production facility broken down by shift (morning, evening, night) is to be plotted. Any of the multiple / clustered histograms could be used (although the clustered varieties might lead to a cluttered plot). An alternative might be to use a multiple boxplot (opposite).

As glance, we can see the morning and evening output is reasonably similar, but the night shift output is lower.

In particular, note that the 3rd quartile for the morning shift approximately aligns with the maximum for the night shift. Thus, in 75% of the shifts considered, output from the morning shift is greater than the maximum output from the night shift – a telling indication that the night shift output is considerably lower.



The various clustered, multiple and stacked charts that we have been looking at can all be applied when we have one scale and one categorical variable. It is often profitable, especially with ready access to chart-building software, to experiment by constructing a number of different charts. It is sometimes the case that a pattern may be very evident in one plot but not in another.

Example

A medical study is undertaken into the treatment high blood pressure in adults. There are to be three aspects to the study.

1. Compare a new drug (A) and an existing one (B). Is there a difference between A and B in the amount by which a patients BP reduces one hour after taking the drug?
2. Does the effect of the new drug kick in as early as 30 minutes after been taken?
3. Does the amount by which the new drug reduces BP vary with patient's age?

Q1

For the first question we would consider a clustered plot (often the boxplot is preferred). We have two groups (new drug/old drug) and the first patient in the old drug sample is not linked or associated with the first person in the new drug sample. Hence the groups are independent.

Q2

Here we could use a difference plot. Again, we have two samples defined by when the BP measure is taken (time_zero, time_30_min). But here, there clearly is a link between the first value in our time_zero group and the first value in our time_30_min group – it is the same patient!

Q3

A scatter plot would be employed here. Two completely different things (BP reduction and age) are measured to address the question – is a patient's age a factor in how effective the new drug is?

Clustered bar chart and χ^2 test

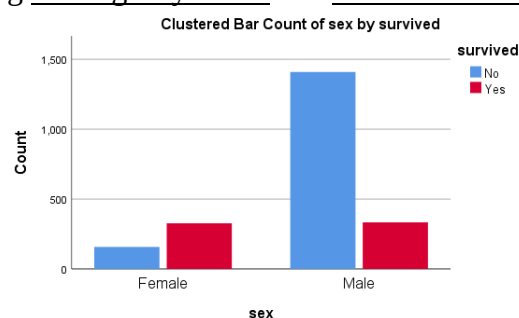
Frequently, especially when social data is being analysed, it will be necessary to examine two categorical variables (remember that categorical variables are needed to represent many quantities in social data – e.g., gender, college course, occupation, nationality, religion, employment status etc.)

(We will look at will be the Titanic data set rather than social data.)

For example, for travellers on the Titanic was there an association between sex and survival? I.e., were the percentages of males who survived different from the percentage of females?

Note that both survival (yes / no) and sex (male / female) are categorical (in fact binary) variables. Categorical variables can often be explored visually using contingency tables and clustered bar charts.

sex * survived Crosstabulation					
			survived		Total
			No	Yes	
sex	Female	Count	158	327	485
		% within sex	32.6%	67.4%	100.0%
	Male	Count	1409	334	1743
		% within sex	80.8%	19.2%	100.0%
Total	Count	1567	661	2228	
	% within sex	70.3%	29.7%	100.0%	



The table on the left is called a contingency table or cross tabulation or cross tab. It clearly shows that women fared much better with about two thirds of their number but only about one fifth of men survived.

Note that there were more male survivors (334) than female survivors (327) because there were far more males than female passengers (about 80%:20%). Somebody examining these data might cite survival numbers rather than survival parents and either unwittingly (or knowingly!) misinterpret them.

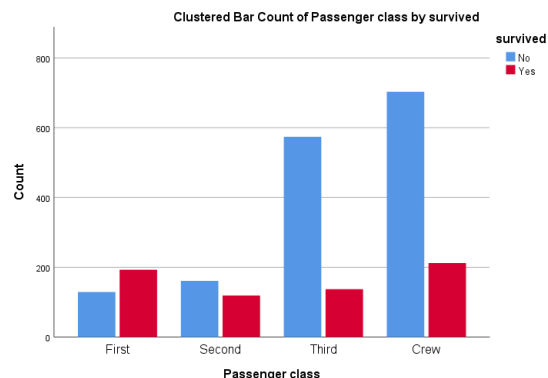
Example

For travellers on the Titanic was there an association between class and survival? I.e., did the percentages of those that survived vary by passenger class?

Note that survived is a binary variable but class (1st, 2nd, 3rd) is an ordinal variable. (We have included crew as a level of class – i.e., travelling as a part of the ship's crew is less luxurious than travelling 3rd class.)

The cross tab and clustered bar chart are again effective at conveying the association.

Passenger class * survived Crosstabulation					
			survived		Total
			No	Yes	
Passenger class	First	Count	129	193	322
		% within Passenger class	40.1%	59.9%	100.0%
Second	Count	161	119	280	
	% within Passenger class	57.5%	42.5%	100.0%	
Third	Count	574	137	711	
	% within Passenger class	80.7%	19.3%	100.0%	
Crew	Count	703	212	915	
	% within Passenger class	76.8%	23.2%	100.0%	
Total	Count	1567	661	2228	
	% within Passenger class	70.3%	29.7%	100.0%	



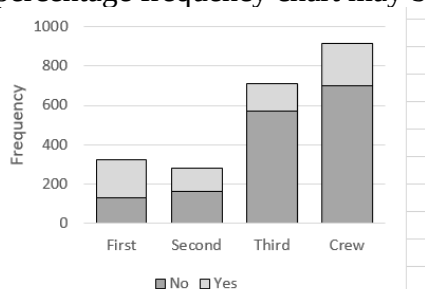
Note that almost 60% of first-class passengers, 43% of second class and 23% of third classes survived. Thus, there is a strong indication that survival prospects diminished.

A suitable test that could be considered here is the χ^2 test of independence. And we can make predictions that we might interpret as probabilities.

For example, if we randomly consider a passenger and want to know the probability that they survived. If the passenger was a first-class passenger, we might take our probability to be 59.9% but revise this down to 42.5% for a second-class passenger etc.

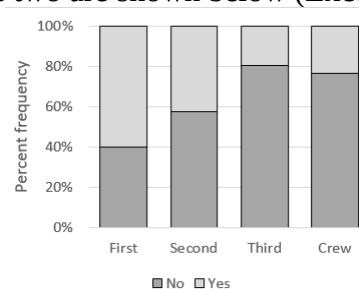
Stacked bar chart

An alternative way to represent the data in the last example is to use a stacked bar chart. Either a frequency chart or a percentage frequency chart may be used. Examples of the two are shown below (Excel)



Stacked frequency chart

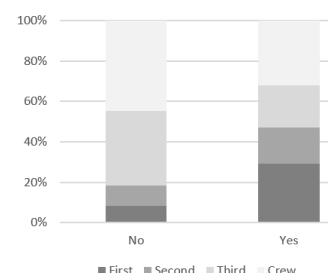
Conveys the numbers in each bar/ class but it is not so easy to compare the survival rate in each bar.



Stacked percentage frequency chart

Easy to compare the survival rate in each bar /class but does not convey the numbers in each class.

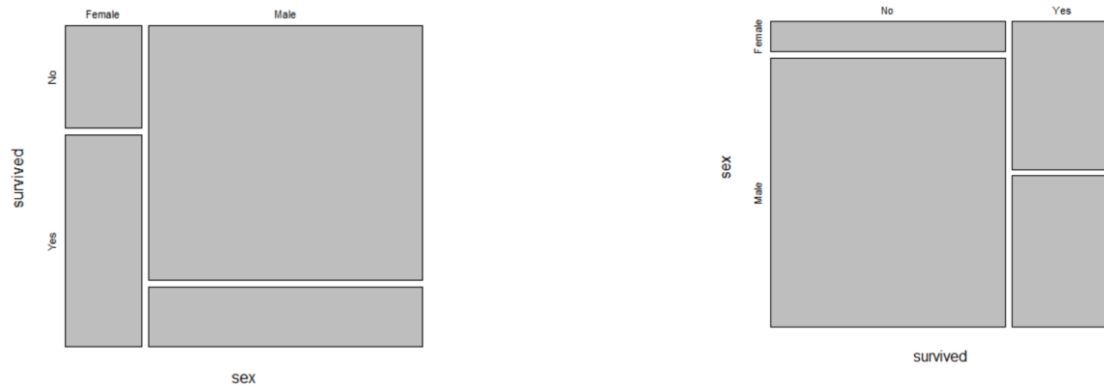
Note that in the stacked plots above, the bars were defined by the levels of class and the two shades within each bar represented the two levels of survived, yes and no. We might alternatively have swapped the roles of the variables, which would lead to the stacked frequency plot shown opposite. This chart aids us in understanding the breakdown of the two levels of survived (yes /no) by the levels of class and is not as effective at showing that as passenger class “increased” from 3rd to 1st, survival propensity diminished.



Thus, the order of selection of the variables very much impacts the look (and consequently, usefulness) of the chart produced. This is true for any plot involving two (or more) categorical variables. This point is illustrated again with the mosaic plot below.

Mosaic plot

The mosaic plot (R) is an alternative means for plotting two categorical variables. The variables *sex* and *survived* for the Titanic dataset are plotted below – and again with the variable order reversed.



The plot on the left is arguably the more informative one. For example, it conveys that male travellers substantially outnumbered female passengers and also, that the male survival rate was much lower.

The plot on the right conveys that about twice as many people did not survive as did, and that those that did not survive were predominantly male, while those that did survive were about equally split between male and female.